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Prepared by: Prof. Arjun Chavada, DU

Unit – 2 ➡ Ecology & Ecosystem

Introduction to Ecology

Introduction:

- The word ecology is derived from two **Greek** words, “**oikos**” meaning house, habitation, or place of living, and “**logos**” meaning study.
- Ecology studies the interrelationship between living organisms and their physical and biological environment.
 - Physical environment includes light and heat or solar radiation, moisture, wind, oxygen, and carbon dioxide nutrients in the soil, water, and atmosphere.
 - Biological environment includes organisms of all kinds, plants, and animals.
- In other words, Ecology means the study of living organisms, both plants and animals in their natural surroundings.

Objectives of Ecology:

The main objectives of Ecological study are to study;

- The interrelationship between organisms in a population and diverse communities,
- The temporal (sequential, timely) changes (seasonal, annual, etc.),
- Understand ecological levels of an ecosystem
- The behavior of organism under natural conditions,
- Understand effects of natural calamities, pollution, urbanization,
- The structural adaptation and functional adjustments of organisms to their physical environment,
- The biological productivity and energy flow in natural systems,

Classification of Ecology:

Ecology may be divided into

- Autecology: it deals with the study of an individual organism or an individual species.
 - In other words, it is the study of the interrelationship between individual species or their population and environment.
 - For example:

- Tree in a forest.
- Synecology: It deals with the study of a group of organisms or species that are associated together as a unit.
- It is concerned with structure, nature, development, and causes of distribution of communities.
- For example:
 - Study of a forest (consisting of various trees, animals, and different organisms)

Based on habitats ecology can be divided as,

- Aquatic ecology: organisms residing in water are included in aquatic ecology, which are marine ecology, freshwater ecology, stream ecology, etc.
- Terrestrial ecology: Organisms residing on land are included in terrestrial ecology, which comprises grassland ecology, desert ecology, and forest ecology.

Classification of Ecosystems

Classification of ecosystems:

→ Based on the habitat ecosystem can be;

(1) Terrestrial ecosystem

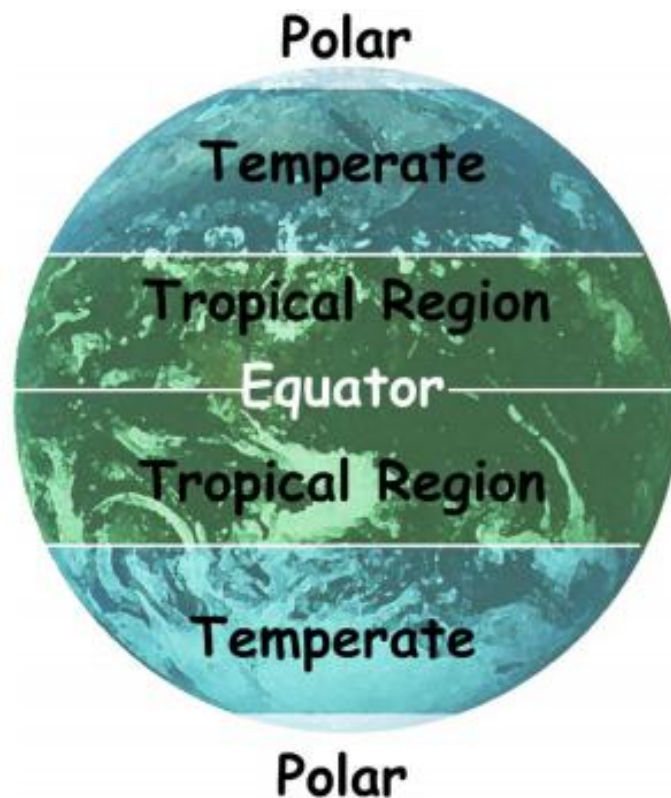
- a) Forest ecosystem
- b) Grassland ecosystem
- c) Desert ecosystem

(2) Aquatic ecosystem

- a) River ecosystem
- b) Lake ecosystem
- c) Ocean ecosystem
- d) Estuarian ecosystem

Forest ecosystem:

→ Forest ecosystems have a predominance of trees that are interspersed with a large number of species of herbs, shrubs climbers, a variety of wild animals, and birds.



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- Depending upon climatic conditions forests can be classified as;
 - a) Tropical rainforest
 - b) Tropical deciduous forest
 - c) Tropical scrub forest
 - d) Temperate rainforest
 - e) Temperate deciduous forest
 - f) Evergreen coniferous forest
- Forests have higher gross production and net ecosystem production compared to any other ecosystem because of their large biomass.
- Hence forests are considered as biotic reservoirs of nutrients.
- Functions of a forest ecosystem:
 - Forest ecosystem enhances water resources in both quantity and quality because of their storage capacity and ecological cleaning action
 - Give shelter to wildlife
 - Acts as a sink for carbon dioxide content in the atmosphere.

Grassland ecosystem:

- Grassland is dominated by grass species but sometimes also allows the growth of a few trees and shrubs.
- **Rainfall is average but erratic (No regular pattern).**
- Limited grazing helps preserve the quality of the soil & food production, but overgrazing leads to the degradation of these grasslands resulting in desert.
- Types of grassland are found based on different climatic conditions;
 - a) Temperate grassland (**25-100 cm** annual precipitation)
 - b) Tropical grassland (**up to 150 cm** annual precipitation)
 - c) Polar grassland: severe cold, strong winds along with ice
- Grassland soils are highly fertile, because of no excessive leaching.
- Grasslands are more prone to erosion.

Desert ecosystem:

- These ecosystems occur in a region where evaporation exceeds precipitation.

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- Precipitation is less than 25 cm a year.
- Deserts have little species diversities and consist of drought-resistant plants.
- Species like, camel & cactus has the water conservation adaptability for their survival in the desert ecosystem.
- The atmosphere is very dry, it has poor insulation, which is why it is hot in daytime and cold in nighttime.
- Deserts are divided based on climatic conditions,
 - a) Tropical deserts:
 - Sahara, Namibia, Thar.
 - Driest desert with only a few species, a wind-blown dune is very common.
 - b) Temperate deserts
 - Mojave in South Africa.
 - High temperature in summer, but cool in winter.
 - c) Cold deserts
 - Gobi Desert in China has cold winters and warm summers.

Aquatic ecosystem:

- based on salinity, aquatic ecosystems can be,
 - (1) Freshwater ecosystem:
 - Freshwater ecosystem can be further divided into 2 parts,
 - a) **Lentic: Standing water** (Lakes, ponds)
 - b) **Lotic: Running water** (Rivers, streams)

Pond ecosystem:

- It is a self-sufficient and self-regulating ecosystem.
- Location, size, depth, and subsurface influence the biology of the pond ecosystem.
- Greatly affected by anthropogenic activity (Human activity).

Lake ecosystem:

- Lakes are usually big freshwater bodies with standing water.
- Lakes have been found to exhibit distinct zones of biological activity, which is determined by the availability of light and oxygen.

→ Important biological zones are,

a) Littoral zone:

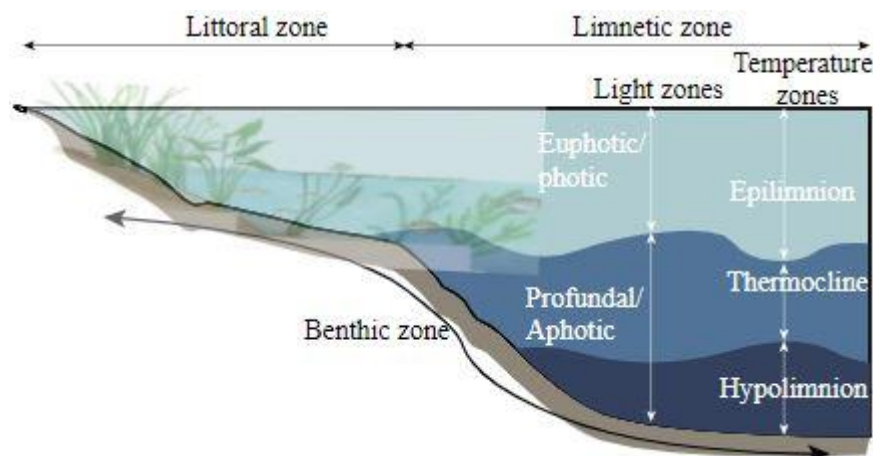
- The shallow water near the shores in which plants are observed.
- This zone cannot extend deeper than the euphotic zone.

b) Euphotic zone:

- The upper layer of water through which sunlight can penetrate is known as the euphotic zone.
- All plant growth occurs in this zone.
- Deepwater algae grow as the main plant.

c) Benthic zone:

- The bottom sediments in a lake comprise this zone.
- As organisms living in overlying layers die, they settle down to the bottom, where they are decomposed by the organisms living in the benthic zone.



→ Stratification:

→ Lakes show stratification based on temperature difference.

→ During summer time the top water becomes warmer than the bottom waters, therefore three distinct layers are created,

a) Epilimnion: warm, lighter, circulating surface water.

b) Hypolimnion: Cold, non-circulating bottom layer.

c) Thermocline: in between two layers, there is a sharp drop in the temperature.

River ecosystem:

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- Rivers are large streams that flow downwards from the mountain highlands and flow through plain falls into the ocean.
- The river ecosystem shows a series of different conditions.
- The mountain highland part has clear cold water rushing down, as waterfalls with large amounts of dissolved oxygen.
- In the second phase on the gentle slopes, the water gets warmer and supports the growth of plants and less oxygen-requiring fishes.
- In the third phase, the river waters are very rich in biotic diversity (various living organisms), they bring with them lots of silt-rich nutrients and form a delta before reaching the ocean.

(2) Marine ecosystem:

- It is composed of an **ocean ecosystem** and an **estuarine ecosystem**.

Ocean ecosystem:

- It covers about 70% of the earth's surface.
- It is the largest, most diverse, and most stable ecosystem of all.
- It has more than 2,50,000 species.
- It is a major sink/reservoir of carbon dioxide.
- Oceans have 2 major life zones; (1) Coastal zone (2) Open-sea

(1) Coastal zone:

- It has relatively warm, nutrient-rich shallow water.
- Due to high nutrients and ample sunlight this is the zone of high primary productivity.

(2) Open sea:

- It is the deeper part of the ocean, away from the continental shelf.
- It is divided vertically into three regions.

a) Euphotic zone:

- Zone which receives abundant light and shows high photosynthetic activity.
- It hosts many species of fish, plankton, marine mammals, etc.

b) Benthic zone:

- It is host to silt, sand, and slowly decomposing organisms.
- This area is very cold due to its depth, where light cannot reach.

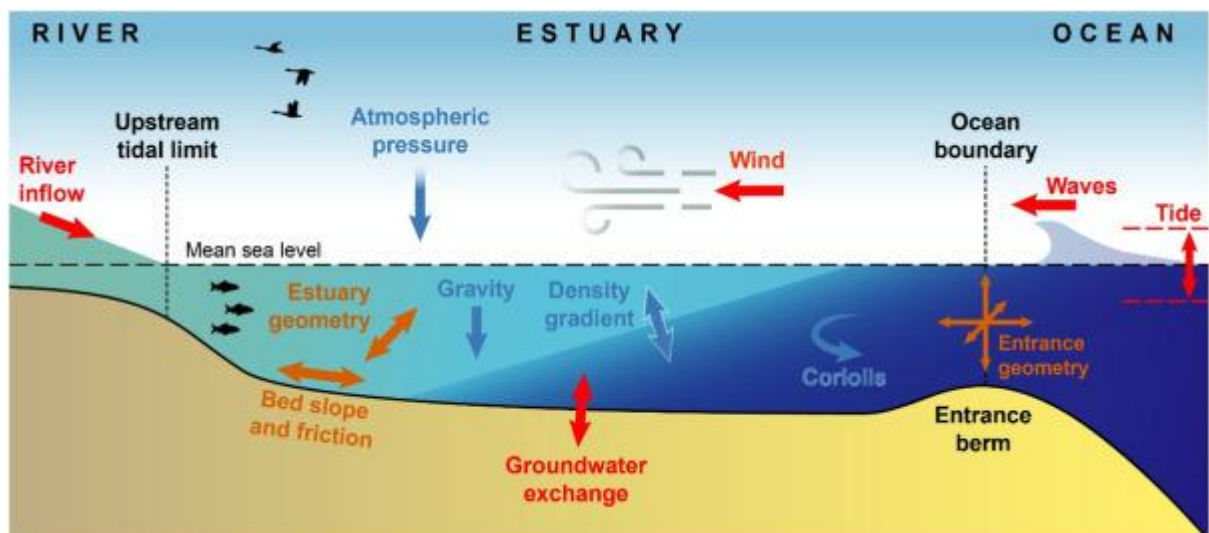
- There are a few plants and animals which include bottom-feeding organisms such as starfish, sponges, etc.

c) Abyssal zone:

- It is the deepest part of the ocean.
- It is the world's largest ecological unit but it is incomplete.
- It has no primary source of energy.
- It is very cold and pressurized.
- Plate movement forms vents in the bottom releasing H_2S which is used by bacteria for survival, and the food chain proceeds.

Estuarine ecosystem:

- An estuary is a semi-closed coastal water body of water that has a free connection with the ocean.
- It is strongly affected by tidal actions, and within its region, fresh water and saline water are mixed.
- The estuaries have rich biodiversity and many species are endemic.
- Migratory species like salmon and eels are also observed which lives half of its life is spent in fresh water and half in saline water.
- Estuary is an ideal place for migratory species to reside.



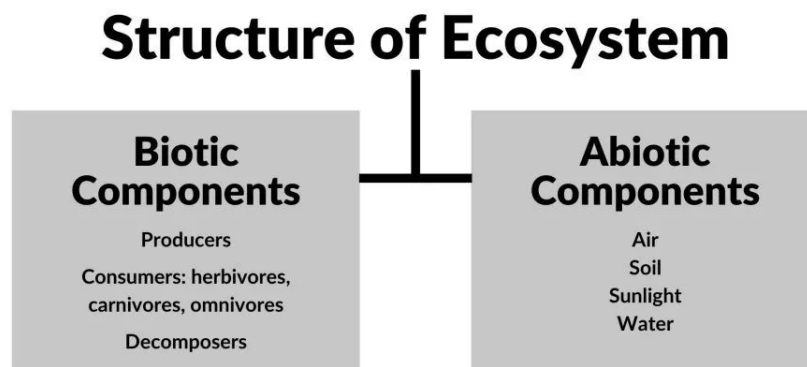
Concept of an Ecosystem

Ecosystem:

- It is made up of two words “eco” and “system”, eco means environment and system implying an interacting and interdependent complex.
- Ecosystem is defined as a **community of interdependent organisms together** with the environment.
- An ecosystem may be as small as a drop of pond water or as large as an ocean.
- It may be temporary, For example:
 - A field of cultivated crops
- It can be permanent, For example:
 - A forest (comprising various organisms) or Ocean.
- An aquarium can be called an artificially established self-sustained ecosystem. Earth itself is a vast ecosystem where biotic and abiotic components are constantly acting and reacting upon each other.

Structure of Ecosystems:

- All ecosystems, whether terrestrial, freshwater, marine, or man-made, have two major components.
- (3) Abiotic components (Non-living): factors such as rain light/sunlight, wind, temperature, soil, etc comes under this component.
- (4) Biotic components (Living): factors such as Animals, birds, fishes, human, and every other organism comes under this component.



Abiotic component:

→ The non-living factors or the physical environment prevailing in the ecosystem form the abiotic component.

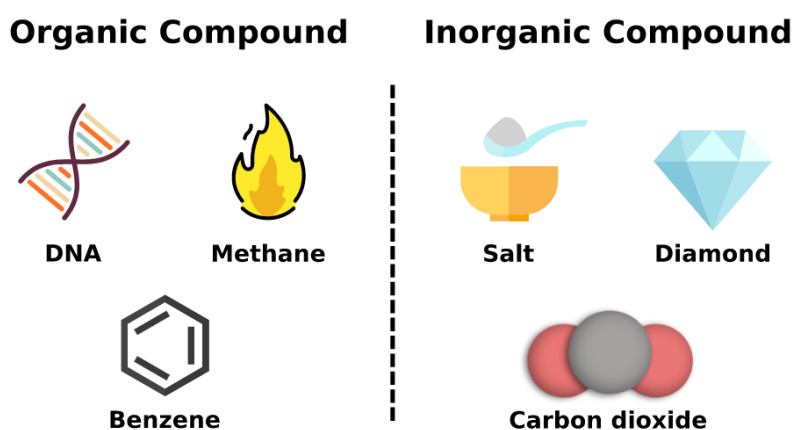
(1) Physical factors which include climatic factors like rain, wind, humidity, light, and wind precipitation, and edaphic factors such as soil, pH, Topography, etc.

(2) Inorganic substances which include water, mineral, and gases. Non-Carbohydrate substances can be included in this division.



(3) Organic substances: Carbohydrates, protein, lipid, and fat, are covered in this division.

→ Abiotic component helps in the survival of the organism residing in the premises.



Biotic component:

→ Living components are divided as below,

(1) Producers: green plants have chlorophyll with the help of which they trap solar energy and change it into chemical energy of carbohydrates using simple inorganic compounds mainly water and carbon dioxide. This process is called "**Photosynthesis**"

→ As green plant manufactures their food on their own they are also called **“Autotrophs”**

→ Part of this energy is used by plants for respiration and some part gets scattered in the environment.

(2) Consumers: the animals which lack chlorophyll are unable to synthesize their food, so they are depended on other organisms for food and survival.

→ They are called **“heterotrophs”**, because of this dependency on others for food.

→ The consumers are of mainly 4 types:

d) Primary consumers or First order consumers or Herbivores:

- These animals feed on plants or producers.
- They are called **“herbivores”**.
- For example:
 - Rabbit, deer, goat, etc.

e) Secondary consumers or Second order consumers or Primary carnivores:

- The animals which feed on herbivores are called primary carnivores.
- For example:
 - Cats, foxes, snakes, etc.

f) Tertiary consumers or Third order consumers:

- These are large carnivores that feed on secondary consumers.
- For example:
 - Wolves

g) Quaternary consumers or Fourth order consumers or Omnivores:

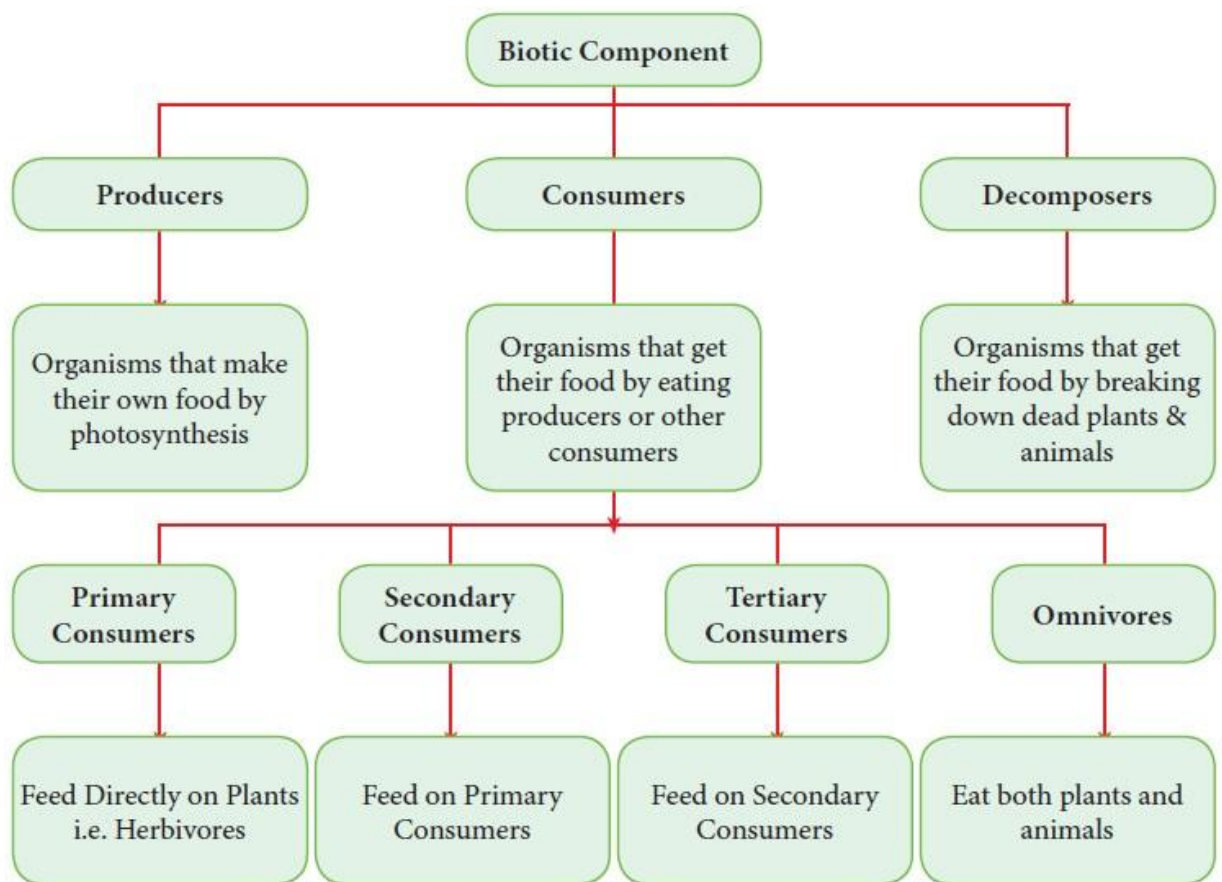
- These are the largest carnivores that feed on the tertiary consumers and are not eaten up by any other animals.
- For example:
 - Lions, tigers.
- **Omnivores** are a kind of animal that eats other animals or plants. (Eaters of everything)

(3) Decomposers or Reducers:

→ Bacteria and fungi belong to this category.

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- They break down the dead organic materials of producers and consumers for their food and release the simple inorganic and organic substances produced as by-products to the environment.
- These substances are reused by producers resulting in a cyclic exchange of material between biotic and abiotic components of the environment.
- The decomposers are also called “**Saprotrophs**”.
- Sapro means “rotten” and trophos means “feeders”.



Biogeochemical Cycle

Introduction:

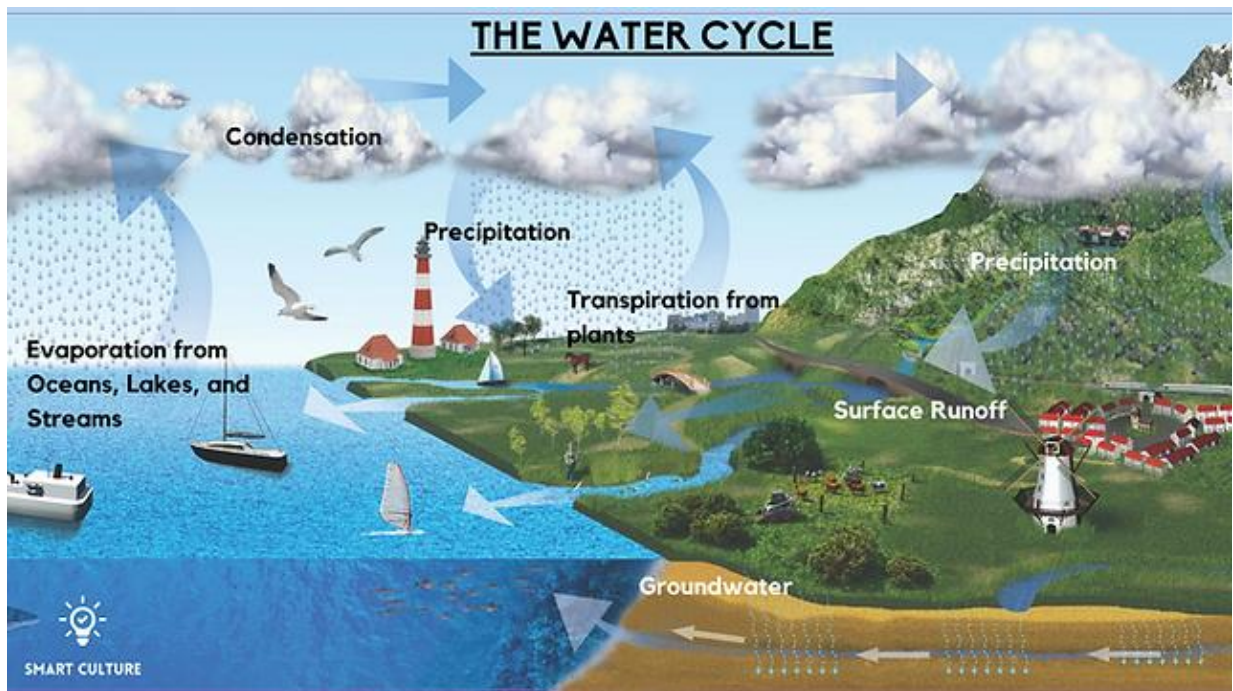
- Nutrients essential for the survival of the organism are in constant circulation in the environment.
- This cyclic translation of particular nutrients in the environment is called the “**Biogeochemical Cycle**”.
- They circulate through life (Bio) and earth (geo) repeatedly (Cycle).
- Biogeochemical includes;
 - Hydrological cycle: movement of water
 - Gaseous cycle: Interchange of oxygen, nitrogen, carbon,
 - Sedimentary cycle: Interchange of minerals (Sulphur, Phosphorus, etc.)

Hydrological or Water cycle:

- **Hydrological cycle** is the process of transfer of moisture from the atmosphere to earth in the form of precipitation, to rivers, and again back into the atmosphere through evaporation.
- The main components of the hydrological cycle are –
 - (1) Evaporation: conversion of water from liquid to gaseous state
 - (2) Transpiration: conversion of water into gaseous form through pores of plant leaves
 - (3) Precipitation: falling of water in liquid (rain) or solid form (hail) from atmosphere to ground
 - (4) Infiltration: water seeping into the soil and getting stored as groundwater
 - (5) Runoff: the flow of water over the ground surface (river)
- The water from the surface sources like lakes, rivers, oceans, etc., gets converted into vapor by evaporation due to solar heat.
- The vapor accumulates continuously in the atmosphere; thus, clouds are formed.
- These clouds cause precipitation (rainfall), and some of the vapor is converted to ice at the peak of the mountains.

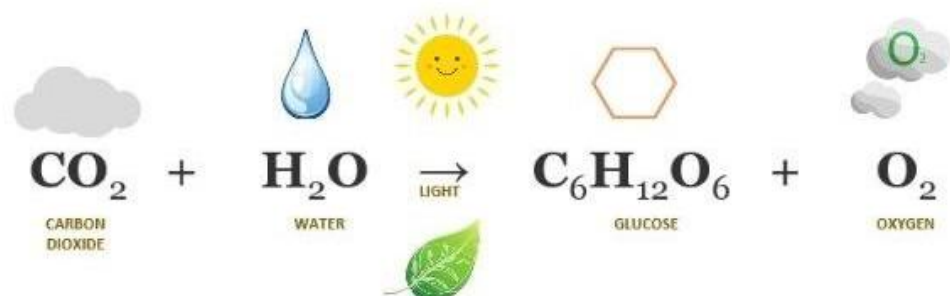
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- These ice caps melt and the water meets the ocean through rivers. Further, some of the water is utilized by plants and vegetation for growth and some is evaporated back into the atmosphere through its leaves, which is known as “**transpiration**”.
- These processes of evaporation, precipitation, transpiration, infiltration, and runoff go on continuously like an endless chain.



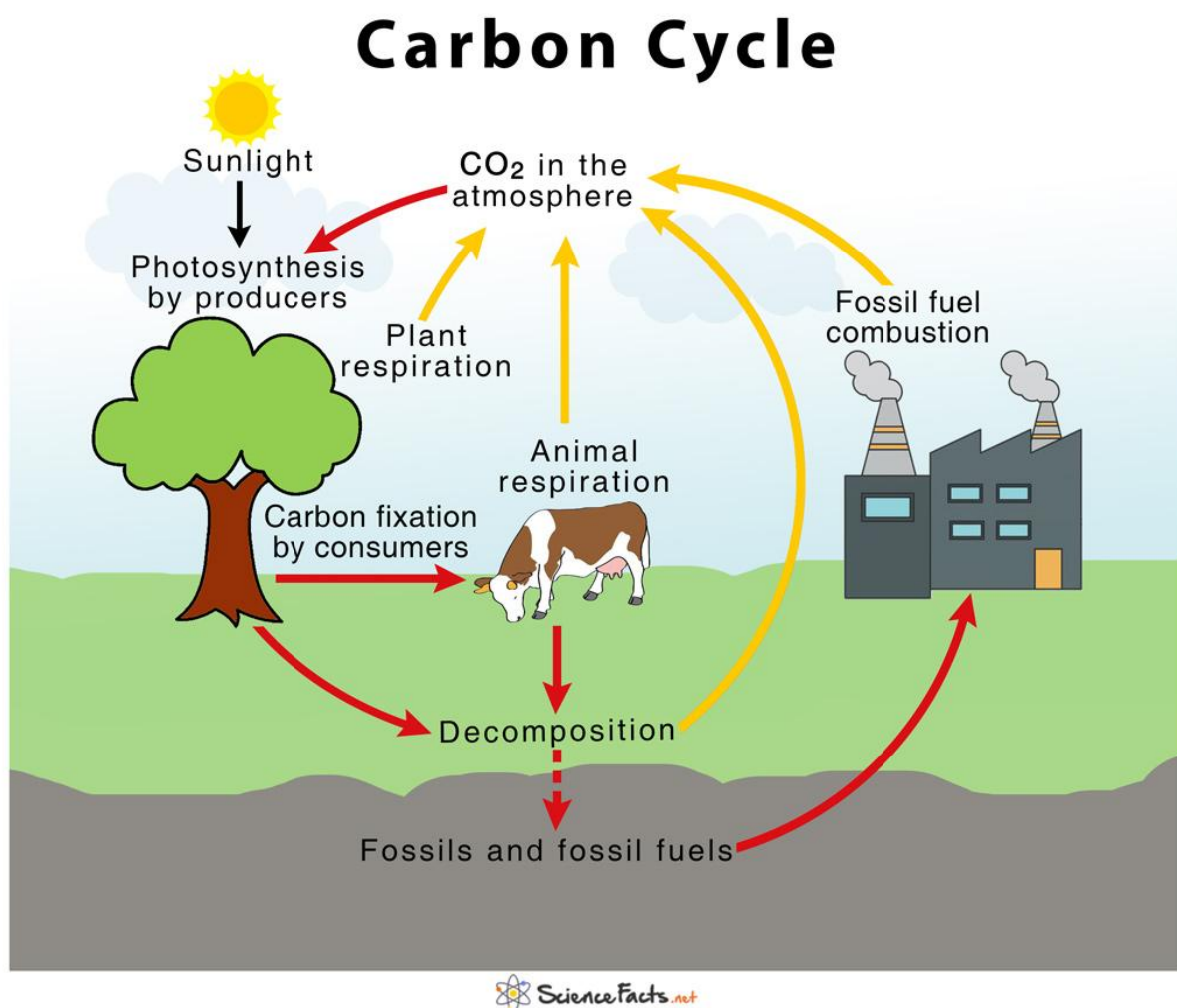
Carbon cycle:

- Carbon in the form of carbon dioxide is taken up by green plants as raw material for photosynthesis, through which a variety of carbohydrates and other organic substances are produced.



PHOTOSYNTHESIS

- The organic compound synthesized is passed from the plants (producers) to consumers (herbivores & carnivores).
- During respiration, plants and animals release carbon back into the atmosphere in the form of **carbon dioxide**.
- The dead bodies of plants and animals as well as the waste which accumulates carbon compounds are decomposed by micro-organisms and release carbon dioxide.
- In addition to the atmosphere, another major reservoir of CO₂ is the ocean. The interchange of CO₂ between the atmosphere and the ocean takes place through diffusion.



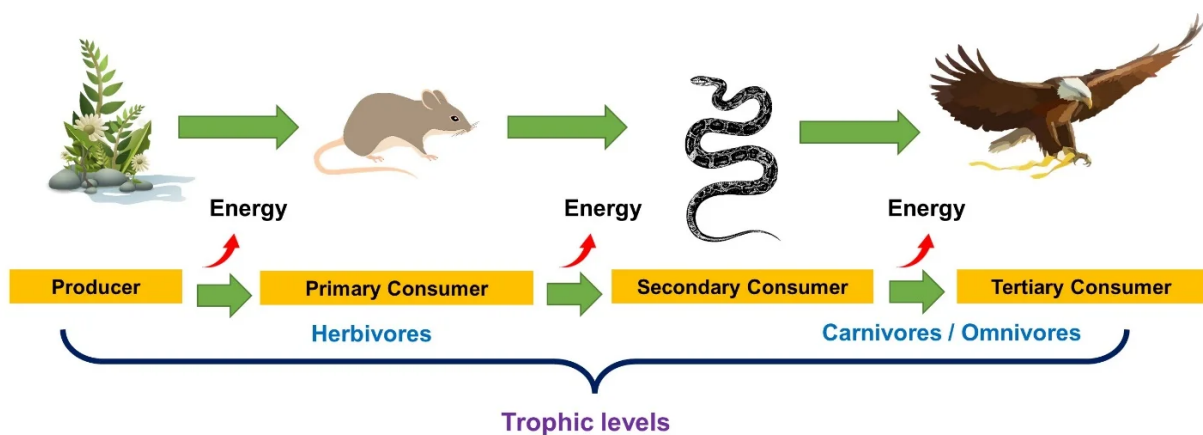
- Carbon dioxide is also recycled during the burning of fossil fuel, in recent years, the carbon dioxide level has increased in the atmosphere due to the burning of fossil fuel which has caused an imbalance in the natural cycle and we are facing issues like global warming.
- Oxygen returns to the surrounding in the form of carbon dioxide or water. It also enters in plant body as carbon dioxide and water during photosynthesis and is released in the form of molecular oxygen as a by-product in the same process.

Food Chain & Food Web

Food chain:

- Graphical representation of organisms eating other organisms, is called a **food chain**.
- In the food chain each organism eats the smaller organism and is eaten by the larger one.
- All those organisms which are interlinked with each other through food constitute the ecosystem.
- The different level in a food chain is called the “**trophic level**”.
- Each food chain has three main trophic levels: **Producer level, Consumer level, and Decomposer level**.
- If any of the intermediate link is removed the succeeding links of the food chain will be affected.

Food chain and trophic levels



- It is estimated that only 10% of the potential energy is transferred from one trophic level to another.
- The efficiency of the food chain is therefore dependent on the number of trophic levels, thus shorter the food chain more is the more energy available.

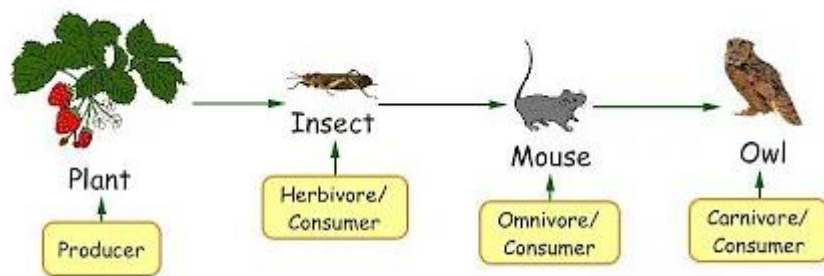
Types of the food chain:

(1) Grazing food chain:

- Such a type of food chain starts with the producer and ends with a higher consumption level.

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→ In every trophic level, the size of the organism increases while its numbers decrease.



A food chain shows the path of energy from one living thing to another.
Decomposers like bacteria, are necessary for all food chains.

(2) Parasitic food chain:

→ This type of food chain starts from a big host and ends with the parasitic organism.

Cow → Worm → Protozoa

(3) Saprophytic food chain:

→ This type of food chain starts from dead organisms and ends with decomposers such as bacteria.

→ It is, however, less efficient as a major portion of the energy is lost to the environment without being properly used.

Dead organism → Fungi → Bacteria

Significance of the food chain:

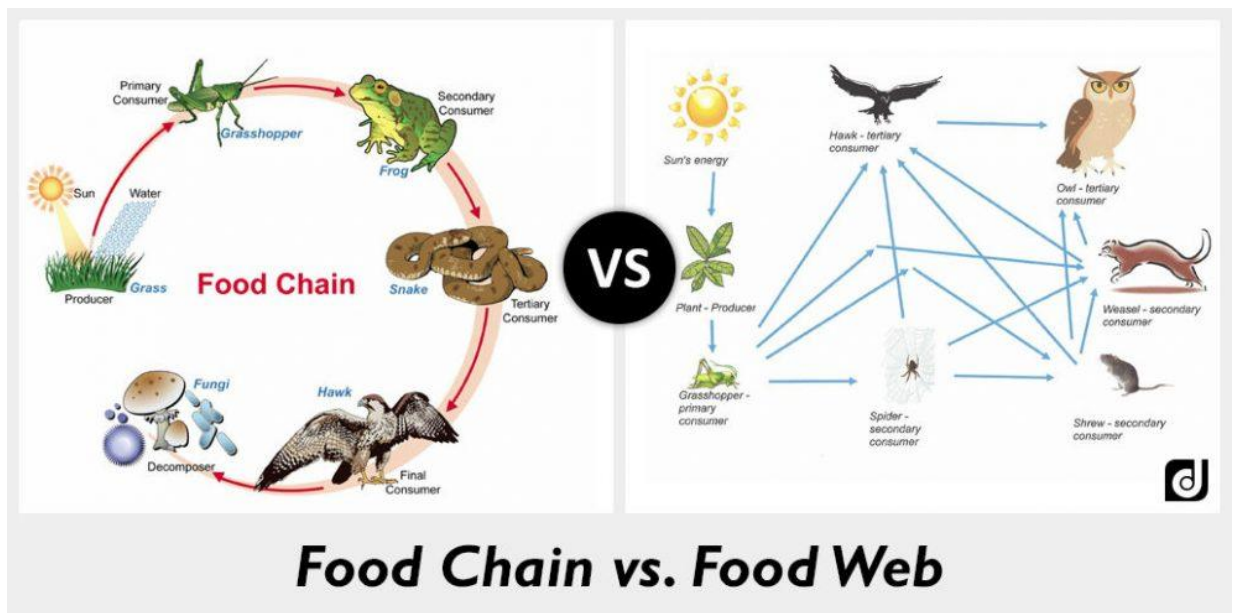
→ To understand the feeding relationship and the interaction between organisms in any ecosystem.

→ To help understand energy flow mechanism and matter circulation

→ Circulation of toxic substances

→ Understand problems of biomagnification

- **Biomagnification** is a tendency of pollutants to get concentrated in successive trophic levels.
- Large concentrations of pollutants could be detrimental if they are toxic.
- Biomagnification occurs when organisms at the bottom of the food chain accumulate toxic material greater than that available in the environment around them.



Energy Flow in an Ecosystem

Energy flow in an ecosystem:

- To maintain life, energy is required. Energy enters an ecosystem from solar radiation.
- In earth's atmosphere about 15×10^8 Cal/m²/year of solar energy is present. Out of which only 47% reaches the earth's surface and only 1 to 5% is converted into chemical energy by green plants.
- Thus, energy from the sun enters the living world through photosynthetic organisms and passes on from one to another in the form of food.
- The flow of energy is unidirectional and non-cyclic.
- Energy enters the ecosystem from solar radiation and is converted into chemical energy by producers, from them energy passes to other consumers.
- This one-way flow of energy is governed by laws of thermodynamics;
 - Energy can neither be created nor be destroyed, it can only be transferred from one to another.
 - During energy transfer there is degradation of energy from a concentrated form (mechanical, chemical, electrical) to a dispersed form.
- No energy transformation is 100% efficient; it is always accompanied by some dispersion or loss of energy in the form of heat.
- Heat energy is not utilized by the biological system and ultimately is lost from the body to the environment.
- It also indicated shorter the food chain, the greater would be the available food energy, and with an increase in the length of the food chain, there is a corresponding increase in loss of energy.

Models for Energy Flow in Ecosystem:

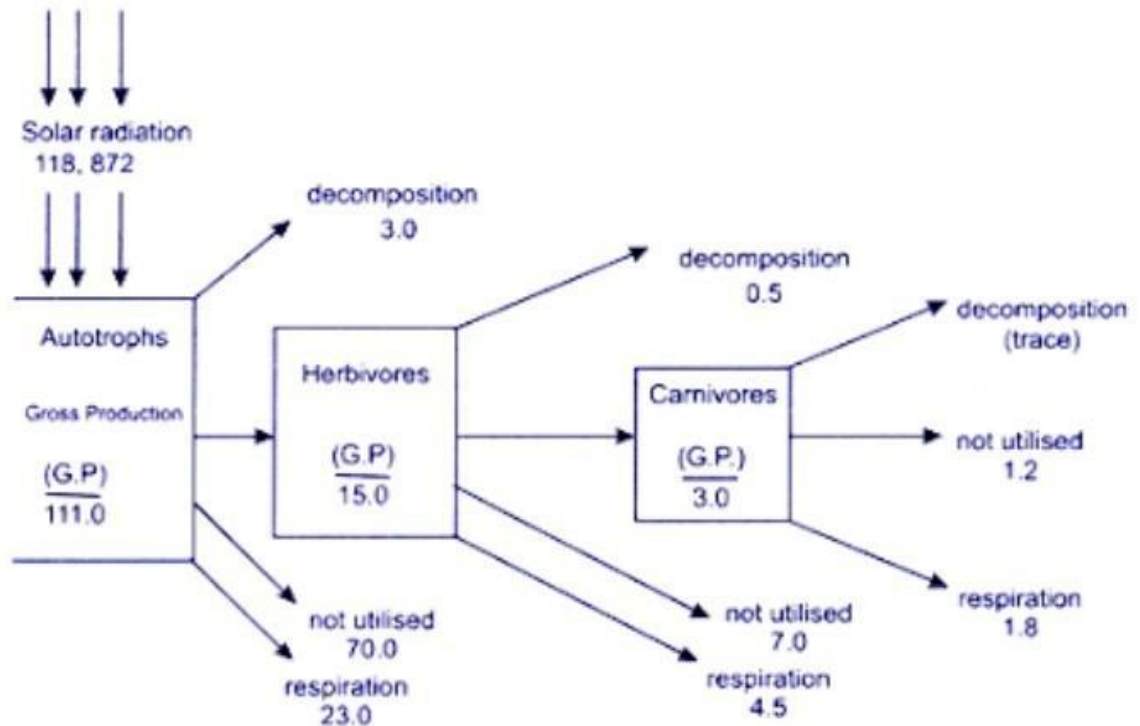
- The flow of energy through various trophic levels in an ecosystem can be explained with the help of various energy flow models:

(1) Single-channel energy flow model:

- The flow of energy takes place unidirectionally through a single channel of producer to herbivores and carnivores.
- The energy captured by autotrophs does not revert to the sun, but passes to herbivores, and that which passes to herbivores does not go back to autotrophs but passes to the secondary consumer.

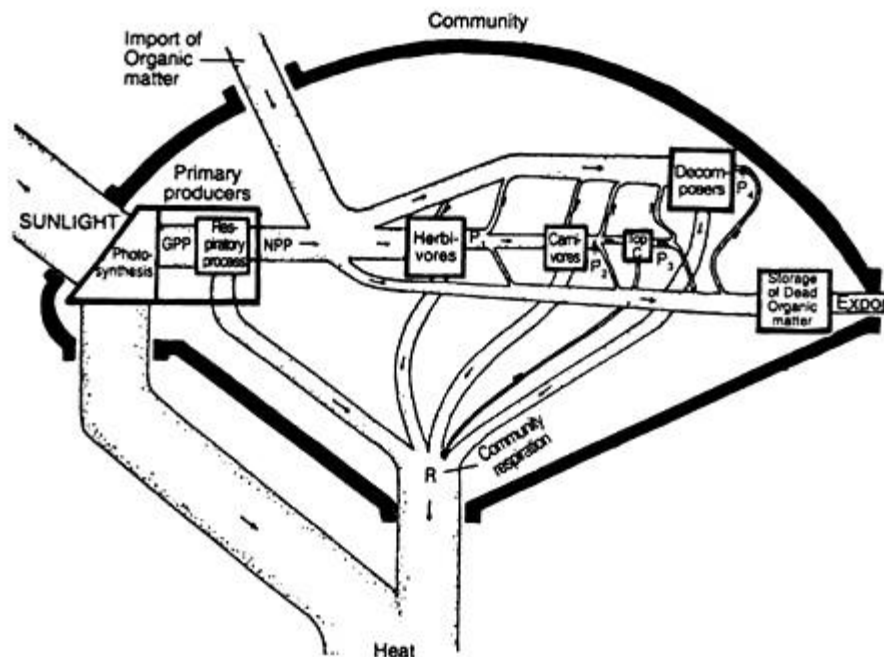
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- Due to this one way of energy flow, if any of the sources of energy is cut off then the entire system would collapse.
- At each trophic level there occurs a progressive decrease in energy which is mainly due to loss of energy in the form of heat.



(2) Y-shaped or double-channel energy flow model:

- Y-shaped energy flow model was introduced by H.T. Odum.
- This model shows common boundary, light, and heat flow as well as import, export, and storage of organic matter.
- Decomposers are separately placed in a box to partially separate grazing and detritus food chains.
- Y-shaped energy flow model is more realistic and practical compared to single-channel model, because of;
 - a) It confirms to the basic stratified structure of an ecosystem.
 - b) It separates grazing and detritus food chains.
 - c) Micro consumer and macro consumer differ greatly in size-metabolism relation in the two models.



(3) Universal energy flow model:

- As the flow of energy occurs, there is gradual loss of energy at each trophic level, resulting in less energy at subsequent trophic levels as indicated by narrower pipes and smaller boxes.

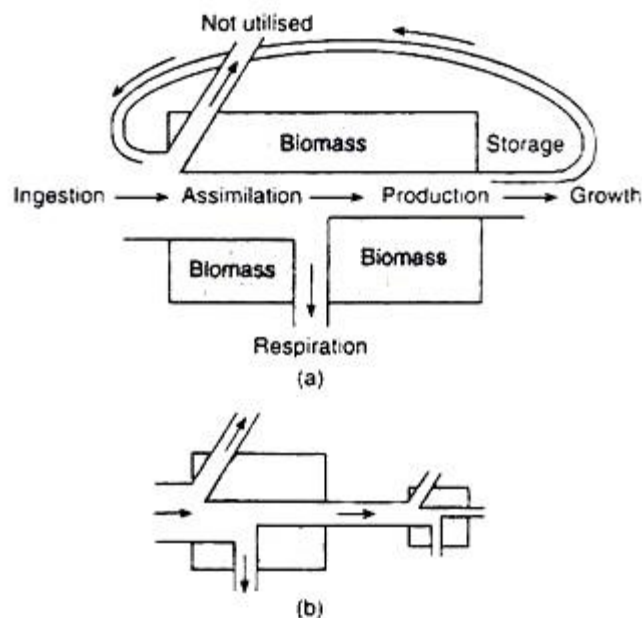
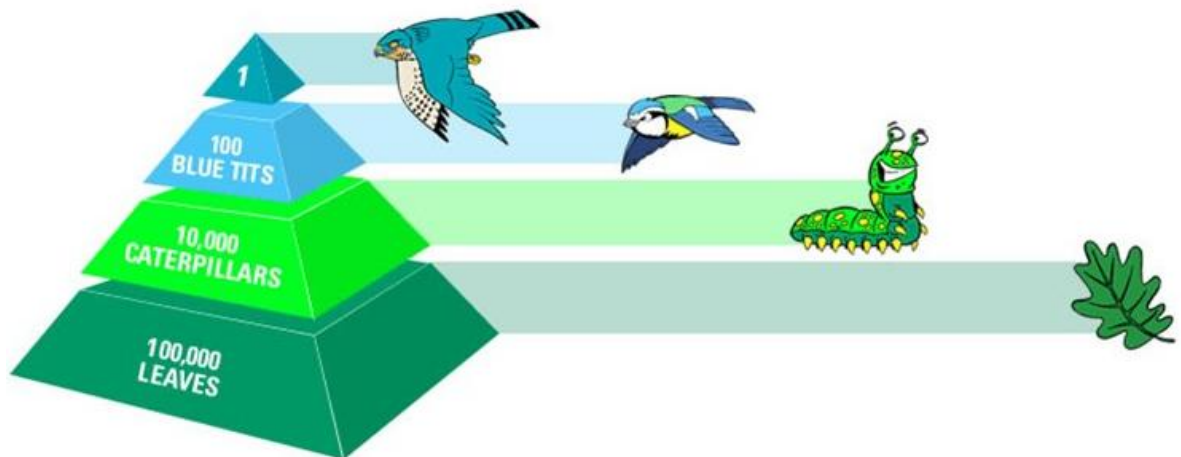


Fig. 4.6 : Universal model of energy flow : (a) Energy flow in one trophic level. (b) A link between trophic levels in a food web

- The loss of energy is mainly energy that has not been utilized (NU).
- Some energy is used for respiration and the rest is used for production (P).

Ecological Pyramids

- Graphical representation of trophic structure and function of ecosystem, starting with the producer at the base and successive trophic levels forming the apex is known as an “**Ecological pyramid**”.



- Ecological pyramids are of three types:

(1) Pyramids of number:

- It represents number of individual organisms at each trophic level of a food chain per unit area at any time (Numbers/m²).
- It can be upright or inverted, depending on the ecosystem and food chain.
- In grassland and aquatic food chains it's upright, but in the parasitic food chains it is inverted.

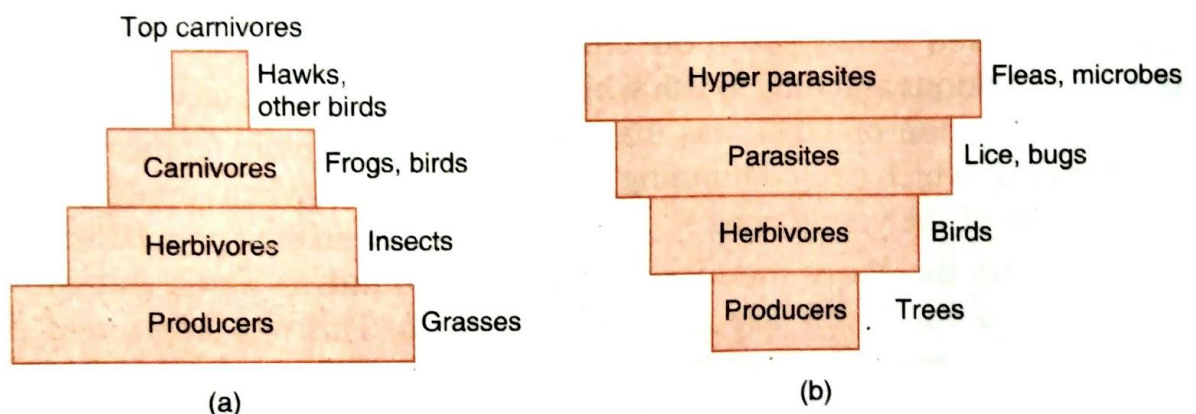


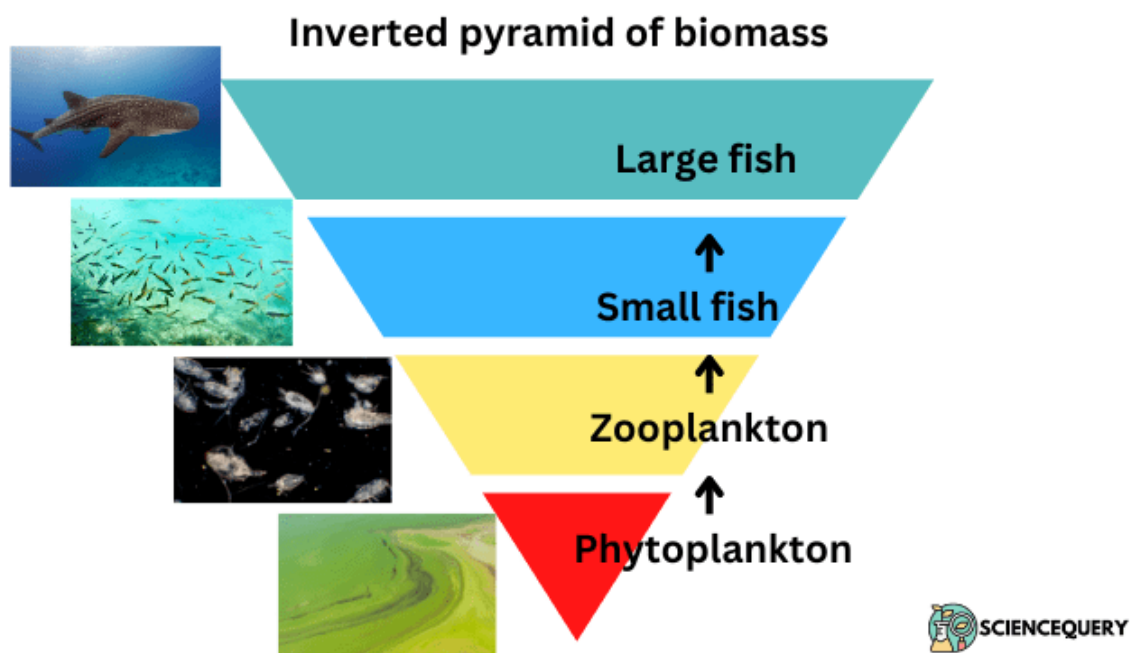
Fig. Pyramid of numbers (a) grassland (b) Parasitic food chain.

- Drawbacks/limitations of the pyramid of numbers;

- Does not take into account the size of the individual.
- The number of individuals of trophic level depends upon biomass.

(2) Pyramid of biomass:

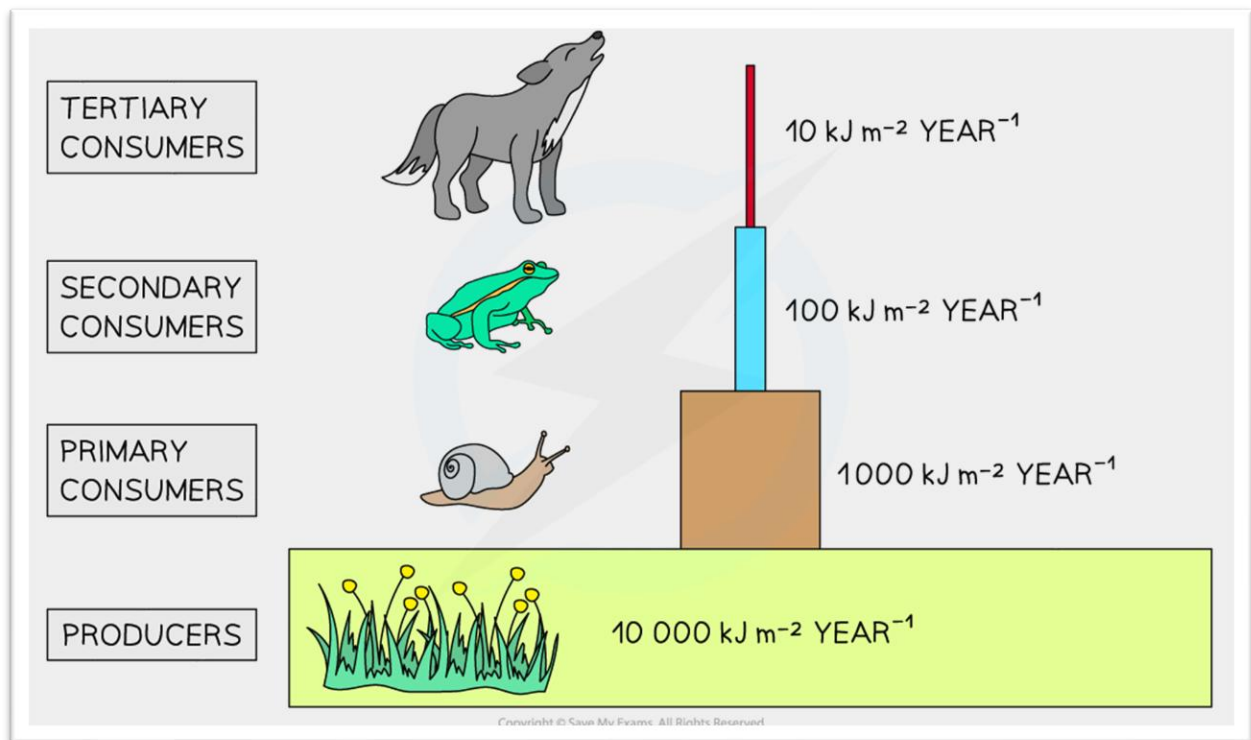
- It represents total biomass (grams/m²) at each trophic level in a food chain.
- It can be inverted or upright.
- In a forest ecosystem the pyramid of biomass is upright, compared to an aquatic ecosystem in which it is inverted.



- Limitation of the pyramid of biomass;
 - Each trophic level indicates standing crops, which are different in every season.
 - Biomass differs in its energy content in different organisms.

(3) Pyramid of energy:

- The amount of energy present in each trophic level is considered for this type of pyramid.
- Pyramid of energy gives the best representation of the trophic level relationship.
- It is always upright, because of less energy at higher trophic levels.
- It is estimated that there is a 10% decrease in the energy from one trophic level to another.



→ Advantages of the pyramid of energy;

- It is always upright.
- It is based on productivity.
- Gives an idea of the actual amount of energy received at each trophic level.